

Welcome to rjf11's Thesis Exercises

AI-Powered Code Intelligence — In-Editor Highlights & Annotations

Before We Begin

Thank you for being here! Your participation is genuinely valuable and helps advance research in how developers learn and work with code.

- Please **remain seated** and commit to the full session so we can gather reliable data.
- Try to **complete each exercise in one go** before moving to the next.
- If you ever need a break, just let Raphael know — no pressure at all.

The Exercises

You'll work through **3 Python exercises**, each in its own folder with a task description, starter file, and reference images.

#	Folder	Topic	Difficulty
1	1 - Charts and Plots	Data Visualization (Seaborn)	Easy
2	2 - Image Manipulation	Image Processing (Pillow)	Medium
3	3 - Network Visualization	Network Graphs (NetworkX)	Hard

You are free to browse the web if needed, though the exercises are designed to be self-contained.

The Tool

You have access to a **code intelligence tool** built into this editor. The goal of this study is to explore whether it helps you feel more productive, understand code better, and stay focused compared to traditional coding or commercial AI tools you may have used.

Selection + Overview

Select any code, then click **"Get Started"**. The tool generates an overview of your selection. From there:

Embed Inline — adds the overview as a code comment

API — function signatures & parameters

Concept — explains what the function does

Usage — how it's typically used

Voice Mode

Toggle the **voice button** on, then speak naturally:

"Select lines 7 to 11"	Select a range
"Select function calc"	Select a function
"Give me an overview"	Get an AI overview
"API" / "Concept"	Get specific details
Any question	Ask anything freely

Gaze Mode

Toggle **gaze mode** on, then simply look at the code. **Look at a function** long enough and it will be automatically selected. **Look at the "Get Started" button** to trigger an overview. A natural, hands-free way to explore and understand code.

Tip: Start each exercise by reading the task `.md` file. The starter `.py` file has scaffolding to get you going, and the reference images show expected output. If anything is unclear, don't hesitate to ask Raphael!

Take a breath, get comfortable, and whenever you're ready...

Open "1 - Charts and Plots" to begin your first exercise

Thank you for your time and contribution.